RESEARCH PROPOSAL

The Desert Thermal Penalty on Water

How Saudi Arabia's Extreme Heat Affects the Cost of Water and Power from a Commercial Heat-Pipe Microreactor — A Coupled Engineering and Economic Study

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Programme: Co-operative Training Programme, 2026

Field: Commercially available microreactors — thermal-hydraulics, systems & techno-economics

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1. Executive Summary

Saudi Arabia faces two linked challenges. First, it makes most of its drinking water by **desalination**, which consumes enormous amounts of energy. Second, much of this water and power is needed in **remote, water-scarce locations** — mines, industrial zones, coastal towns and giga-projects — where extending the national grid is costly. A **microreactor** (a very small, factory-built, truck-deliverable nuclear unit) can supply both electricity and heat for water production at exactly these sites, running for years without refuelling.

There is, however, a catch that no published study has measured. The most suitable microreactor for the desert — the **Westinghouse eVinci** — rejects its waste heat to the **air, not water** (a sensible choice where water is scarce). But Saudi summers reach **45–55 °C**. When the surrounding air is already that hot, getting rid of heat becomes harder, much like a car radiator struggling on a hot day. This **"desert thermal penalty"** slightly lowers how much the reactor can produce and therefore raises the real cost of the water and electricity it delivers.

This project will **measure that penalty for the first time**: it will model how a heat-pipe microreactor behaves when the outside air is 25–55 °C, confirm it stays safe, and then feed that result into an economic model to show how the cost of **water (LCOW)**, **heat (LCOH)** and **electricity (LCOE)** change with temperature — for real Saudi coastal sites, and compared against today's diesel-and-grid alternatives. The whole study uses **free, open-source software and public data**, so it is fully reproducible and achievable by a single student.

2. Background and Why It Matters

Saudi Arabia's civil-nuclear programme is organised under the Saudi National Atomic Energy Project, with a national ambition of roughly 17.6 GWe of nuclear capacity by 2040[1]. Within this programme, **desalination is repeatedly named as a top national application** for nuclear energy: it is written directly into the charter of the **Saudi Nuclear Energy Holding Company (SNE)**, which covers electricity, seawater desalination and thermal applications[2], and **KACARE** has explicitly stated its interest in small reactors for "remote areas, industrial zones, and applications like water desalination and hydrogen production"[3]. The

Ministry of Energy likewise ties the nuclear mandate to desalination and industrial heat[4]. In short, the water–energy link is a stated national priority, not an academic guess.

At the same time, water scarcity makes siting difficult: geographic studies show that the shortage of cooling water disqualifies the large majority of Saudi land from hosting conventional, water-cooled nuclear plants[9]. This is precisely why an **air-cooled microreactor** is attractive — it can be placed where large water-cooled plants cannot. Microreactors therefore appear most often in Saudi studies of remote power and desalination (for example, an eVinci techno-economic study for THE LINE[11]), and the topic is active at the national level — the 2025 SCOPE-2 conference at KFUPM featured hundreds of papers on reactor thermal-hydraulics, desalination and advanced fuels[20].

The piece that is missing is quantitative: although heat-pipe microreactors and dry (air) cooling are each studied separately[12,13], **no peer-reviewed work has quantified how extreme desert ambient temperatures change the safety margin, the output, and ultimately the cost of cogenerated water and power** for a commercial microreactor. That is the gap this proposal fills, and it is the single most decision-relevant unknown for any Saudi microreactor deployment.

3. Research Objectives and Questions

The project answers one central question — **"How much does Saudi Arabia's extreme heat raise the cost of the water and power a microreactor can make, and is it still worth it?"** — through four objectives:

1. **Build** a thermal model of a commercial heat-pipe microreactor and confirm it can safely reject heat and remove decay heat across ambient temperatures of 25–55 °C.
2. **Produce** a "derating curve" — how net output and capacity factor fall as ambient temperature rises.
3. **Build** an economic and dispatch model that turns this thermal behaviour into the cost of water (LCOW), heat (LCOH) and electricity (LCOE) at a real Saudi coastal site.
4. **Compare** the microreactor against diesel-plus-desalination and grid-plus-desalination, and report the result with full uncertainty bands so the conclusion is defensible despite limited cost data.

4. Reference Microreactor Design

To keep the study grounded in real, commercially available technology rather than a hypothetical core, it is anchored to published designs.

Primary reference — Westinghouse eVinci. A heat-pipe-cooled microreactor of about **13 MW-thermal / 5 MW-electric**, **air-cooled and requiring no water for cooling**, with a compact footprint and a sealed core lasting 8+ years[5,6]. Its water-free cooling is exactly why it suits arid Saudi siting.

Comparator — USNC Micro-Modular Reactor (MMR). A high-temperature (HTGR) microreactor with a molten-salt heat store whose high outlet temperature favours thermal (MED) desalination[7]. It is used as a higher-temperature comparison case.

Scale baselines — the Korean **SMART100** reactor (a desalination-capable SMR)[8], and the conventional **diesel-generator and grid** supplies that microreactors would displace.

5. Methodology — How the Research Will Be Done

The work is deliberately **phased** so that a complete, publishable result exists early and the harder modelling builds on it. The two phases share the same economic model, so no effort is wasted.

Phase 1 (Months 1–6) — The economic spine

Build the economic and dispatch model first, because it relies on the most readily available data. For a chosen Saudi coastal site (e.g. Yanbu, Duba or a Gulf industrial node), assemble an hourly demand profile for electricity and water. Model the microreactor together with reverse-osmosis (RO) and thermal (MED) desalination, solar PV, battery and grid connection, and let the model decide hour-by-hour whether to **sell power or make water**. Desalination performance and cost are taken from the IAEA DEEP tool[17] and calibrated to real Saudi figures — the Saline Water Conversion Corporation recently reported costs as low as SAR 1.7/m^3[10]. Because microreactor cost data are still uncertain, every cost input is entered as a **range, not a single number**, and results are reported as probability bands. This phase is a complete paper on its own.

Phase 2 (Months 5–11) — Adding the desert thermal penalty (the core novelty)

Model the reactor's heat behaviour and feed it into the Phase-1 economics. The chain has four simple steps:

- **Heat source:** a reactor-physics model (OpenMC) produces the decay-heat the reactor must remove, validated against the KRUSTY experiment — the only nuclear-tested heat-pipe reactor[16,15].
- **Cooling in the heat:** a heat-transfer model (OpenFOAM) of the air-cooled path is run at 25, 35, 45 and 55 °C ambient; a safety model (MELCOR) confirms the reactor stays within safe temperatures if cooling is lost[14].
- **The penalty curve:** these give net efficiency and capacity factor as a function of ambient temperature — the derating curve.
- **The link:** the derating curve is fed back into the Phase-1 economic model, so a hot afternoon lowers output and shifts the water-versus-power decision, changing the cost of water and power. This engineering-to-economics link is the part nobody has published for microreactors.

Throughout, the modelling uses only **open, freely available tools** — OpenMC, OpenFOAM, MELCOR and the HERON/RAVEN systems-and-uncertainty toolkit[16,18] — avoiding any licence-restricted software. The thermal results can optionally be contributed to the international OECD-NEA heat-pipe microreactor benchmark[19], which is itself a recognised publication route.

Key data sources

What is needed	Primary source	Availability
Reactor specs & geometry	eVinci / MMR public data; INL open heat-pipe-microreactor reference plant[5,7,14]	Open / public
Desalination cost & performance	IAEA DEEP tool; SWCC reported costs[17,10]	Free / public

What is needed	Primary source	Availability
Climate & seawater temperature	KACARE atlas; NASA POWER; Copernicus Marine	Free / public
Thermal & physics validation	KRUSTY benchmark; code-to-code checks[15]	Open
Software	OpenMC, OpenFOAM, MELCOR, HERON/RAVEN[16,18]	Open-source

6. Alignment with Saudi Vision 2030 and National Priorities

- Directly serves the **water-security and desalination** priority named by KACARE, the Ministry of Energy and written into the SNE charter[2,3,4].
- Provides KACARE, SNE and the regulator (NRRC) with the **first quantitative answer** on whether microreactors pay off in Saudi conditions — informing the large-versus-small reactor decision.
- Supports **localisation**: an original, Saudi-specific framework and an open, reusable tool that builds domestic capability rather than importing a foreign study.
- Unlocks the remote, arid sites — mines, industrial zones, coastal towns — that **cannot host large water-cooled plants**[9].

7. Expected Deliverables and Publication Targets

- **Paper 1 (Phase 1):** micro-scale desalination/cogeneration techno-economics for a Saudi coastal site — target **Desalination** or **Energy Conversion and Management** (Q1).
- **Paper 2 (Phase 2, flagship):** the desert thermal penalty coupled to cogeneration economics — target **Energy Conversion and Management** or **Applied Energy** (Q1).
- A documented, reusable **open-source Python/HERON pipeline** and the ambient derating maps, published with a DOI.
- Optional contribution to the **OECD-NEA heat-pipe microreactor benchmark**[19].

8. Proposed Timeline

Period	Focus	Key activities & milestone
Months 1–3	Setup & data	Literature review; choose site; gather climate, demand, cost and desalination data.
Months 3–6	Phase 1 — economics	Build dispatch + LCOW/LCOH/LCOE model with uncertainty; write Paper 1. ◆ Submit Paper 1.
Months 5–9	Phase 2 — thermal	OpenMC decay-heat + KRUSTY validation; OpenFOAM/MELCOR dry-cooling sweep at 25–55 °C; build derating curve.
Months 9–11	Coupling	Feed derating curve into the economic model;

Period	Focus	Key activities & milestone
		full uncertainty analysis; write Paper 2. ◆ Submit Paper 2.
Months 11–12	Wrap-up	Thesis integration; (optional) ML surrogate and OECD-NEA benchmark contribution; defence preparation.

9. Conclusion

This proposal turns a clear national priority — making water and power affordably in a hot, water-scarce country — into a focused, original and achievable research project. By anchoring the study in a real commercial microreactor, relying entirely on open data and software, and reporting results with honest uncertainty, the project is deliverable within the co-op timeline while producing two high-quality (Q1) journal papers and a reusable national decision tool. It answers a question that is both genuinely new in the literature and genuinely useful to KACARE, SNE and the NRRC: **does the desert's heat change whether microreactors are worth it for Saudi Arabia's water and power — and by how much?**

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